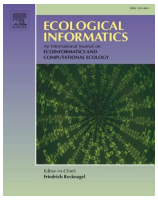
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Kalman filtering assimilated machine learning methods significantly improve the prediction performance of water quality parameters

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ABSTRACT

Accurate water quality prediction is essential for effective water pollution prevention and emergency responses. However, existing research on machine learning (ML)-based data assimilation methods remains limited, particularly in terms of addressing the combined impacts of climate change and anthropogenic activities. To address this gap, we proposed a novel 'ML–Kalman filter (KF)' data assimilation framework and evaluated its performance in the Dahei River Basin, a representative semi-arid watershed. Our results demonstrated significant improvements in predicting key water quality parameters, including total nitrogen (TN), total phosphorus (TP), and the permanganate index (COD_{Mn}), through the integration of KF with four ML models (LSTM, RF, XGBoost, and SVR). The accuracy enhancement ranged from 4.3 % to 17.6 %, with TP showing the most substantial improvement (9.2 %–17.6 %), followed by TN (6.4 %–11.1 %) and COD_{Mn} (4.3 %–12.1 %). After assimilation, the models exhibited the following performance ranking for TN based on the coefficient of determination (R^2): LSTM–KF ($R^2 = 0.909$) > RF–KF ($R^2 = 0.886$) > SVR–KF ($R^2 = 0.840$) > XGBoost–KF ($R^2 = 0.797$), with similar trends observed for TP and COD_{Mn}. The proposed framework demonstrates strong portability and applicability across different monitoring sections and temporal resolutions, offering a robust solution for regions with limited monitoring capabilities and challenging climatic conditions. These findings provide valuable data and technical support for advancing water pollution prediction and early warning systems, particularly for ecological and environmental departments operating in data-deficient regions.

1. Introduction

Water quality is a pivotal component of the water environment system and a direct barometer of overall environmental health (Cha et al., 2017; Wang et al., 2019a; Wu et al., 2016). However, the exacerbation of water quality deterioration driven by climate change and intensified human activity has created an urgent need for advanced monitoring and predictive systems. Therefore, accurate water quality prediction is of paramount importance to ensure water resource security and sustainability, which are fundamental to human health, ecosystem equilibrium, and economic prosperity (Fang et al., 2018; Li et al., 2024; Zhi et al., 2024).

Artificial intelligence has emerged as a transformative tool. Machine learning (ML) methods, including neural networks, support vector machine (SVM), decision tree (DT), and random forest (RF), have demonstrated remarkable efficacy in diverse domains owing to their ability to discern intricate patterns from complex datasets (Hinton and Salakhutdinov, 2006). By combining long-term records of temperature, pH, dissolved oxygen, nutrients, and pollutant concentrations, these models uncover latent relationships that support accurate forecasts. Artificial neural networks (ANNs), for example, have been widely used to estimate biochemical oxygen demand (BOD) in aquatic systems (Barzegar et al., 2016; Goz et al., 2019; Zhang et al., 2019; Zhu et al., 2019). Hybrid schemes that fuse ANNs with remote-sensing data further improve retrieval of water-quality parameters (Anand et al., 2024; Chen et al., 2021a; Hill et al., 2020; Mamun et al., 2024), while ANN–genetic-algorithm ensembles deliver additional accuracy gains (Palani et al., 2008). Other algorithms—including SVM, DT, RF, and adaptive-

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network-based fuzzy inference systems (ANFIS)—have also proved effective in regression tasks for surface-water quality (del Castillo et al., 2024; Jaloree et al., 2014; Najafzadeh et al., 2016; Yajima and Derot, 2018). Deep learning, which is an advanced subset of ML, has revolutionised fields such as image classification, natural language processing, time-series analysis, and applied in the environmental field (LeCun et al., 2015; Li et al., 2017; Piotrowski et al., 2015; Reichstein et al., 2019; Ren et al., 2022; Wang et al., 2019b; Xiang et al., 2020). In water quality prediction, deep learning models, particularly long short-term memory (LSTM) networks, have gained prominence owing to their ability to capture temporal dynamics and process nonlinear relationships in time-series data (He et al., 2022; Saha et al., 2023; Wang et al., 2019b; Zhang et al., 2018; Zhi et al., 2021). LSTM's unique architecture of LSTM cells, which retains and updates long-term dependencies, has proven effective in diverse applications, including sewage plant pollutant prediction, cyanobacterial bloom forecasting, and the estimation of key pollutants, such as total nitrogen (TN), total phosphorus (TP), and dissolved oxygen (DO) (Chen et al., 2021b; Chen et al., 2022; Liu et al., 2022; Prasad et al., 2022; Zhang et al., 2018; Zhang et al., 2022).

Despite these advancements, several inherent drawbacks limit their real-world performance: Sensitivity to sensor noise and outliers, which are pervasive in automatic stations; Vulnerability to missing or irregularly spaced observations caused by maintenance, power loss, or telemetry failure; and Limited physical interpretability and error propagation control when models are deployed in dynamic river systems. These issues often yield biased or highly uncertain forecasts, even when advanced hyper-parameter tuning is applied. Data assimilation methods, such as Kalman filtering (KF), particle filters, and ensemble methods, have been employed to enhance prediction accuracy by integrating observational data with models (Kim and Lee, 2021; Loos et al., 2020). Recent studies have demonstrated the potential of combining ML with data assimilation techniques in fields such as autonomous driving, trajectory correction, regression analysis, and environmental research (Ahmad et al., 2024; Cai et al., 2019; Huang et al., 2020; Wu et al., 2017; Yang et al., 2024). This method has also been used in many studies in water quality prediction. Wang et al. (2023) compared several data assimilation techniques in water quality models for the field of oxygen reaeration and heterotrophic bacteria activity. In order to reduce the noise of input data, RF was combined with KF to improve the water quality prediction accuracy of South Platte River in United States (Poursaeid, 2025). EnKF-ANN was used to predict water quality in two rivers in Malaysia, and by combining the ITD approach, the RMSE of the prediction was reduced by 24 % and 34 % respectively (Rezaie-Balf et al., 2020). The EnKF and smoothing algorithms were also applied in the data fusion system to improve the accuracy of daily chlorophyll-a data along the northern coastline of the Gulf of Finland (Gunia et al., 2022). Some research developed a Kalman filter–based multi-sensor data fusion model for real-time lake water quality assessment, effectively improving prediction accuracy and stability under dynamic environmental conditions (Priyanka et al., 2024).

In this study, we explored the integration of ML models with KF for water quality prediction. Specifically, we proposed several research questions. Which machine learning model, when integrated with the Kalman Filter, demonstrates the highest predictive accuracy for key water quality parameters under noisy and variable monitoring conditions? To what extent can the KF-assisted ML framework be generalized across multiple monitoring sites and temporal resolutions in semi-arid river systems? To answer these questions, we proposed a novel framework that combines four ML models (LSTM, RF, XGBoost, and SVR) with KF. The main objectives of this study were the following: (1) construct water quality prediction models using the LSTM, RF, XGBoost, and SVR algorithms based on water quality evaluation results; (2) apply the KF algorithm in conjunction with these models for data assimilation and assess the improvement in prediction accuracy; and (3) evaluate the prediction performance of the integrated models and their portability across different monitoring sections and time resolutions.

2. Study area and dataset

2.1. The Dahei River Basin

The Dahei River, a representative primary tributary of the middle reaches of the Yellow River, has a length of 238 km and drainage area of 17,673 km^2. Its water system primarily consists of the Dahei River, Xiaohei River, and various tributaries in the western region. It flows through the suburbs of Hohhot City and merges with the Yellow River near Toketuo County, with the mainstream flowing from the northeast to the southwest. Mountainous and loess hilly areas account for approximately 71 % of the basin area, while plains occupy 29 %, making them significant grain bases in Inner Mongolia. It experiences a temperate semi-arid continental climate, with annual precipitation ranging from 330 to 460 mm. Owing to intensified human activities, such as urbanisation and agricultural irrigation, as well as insufficient natural replenishment of river water due to local climatic factors, the water quality status within the Dahei River Basin, particularly in its main stream and tributaries, such as the Xiaohei River, is notably compromised, indicating poor overall conditions. In total, 17 monitoring sections were established: four on the main stream, four on the Xiaohei River, and nine on various tributaries in the western region. Water quality of classes II–III accounted for 58.8 %, Class IV accounted for 11.7 %, and poor quality (Class V) accounted for 23.5 %. The main pollutants included TN, permanganate index (COD_{Mn}), and TP, indicating an overall moderate pollution status of water quality.

Daily water quality monitoring was conducted within the Ruhuangkou section (RHK), located at the terminus of the main stream of the Dahei River, from September 2018 to September 2023. Data were sourced from the local ecological environment management department. Nine parameters were monitored: water temperature (WT), pH, TN, TP, DO, NH_3–N, COD_{Mn}, turbidity (TB), and conductivity (EC). We collected a batch of meteorological and hydrological data corresponding to the daily water quality data. Meteorological data were obtained from the Hohhot meteorological station, including average temperature (tavg), minimum temperature (tmin), maximum temperature (tmax), total precipitation (prcp), snow depth (snow), wind direction (wdir), average wind speed (wspd), and sea level pressure (pres). Hydrological data were obtained from the Sanliang hydrological station, and the index was the flow. The statistical descriptions of all daily scale data are presented in Table 1. (See Fig. 1.)

We also collected monthly monitoring data from nine sections of the main stream of the Dahei River and its tributary, the Xiaohei River, spanning the period from January 2013 to December 2022. In addition to the nine daily monitoring parameters, the dataset includes two additional parameters: chemical oxygen demand and BOD. This monthly scale dataset was primarily used to develop a monthly prediction model for water quality indicators and to evaluate whether the KF could enhance the prediction accuracy. The statistical descriptions of all monthly scale data are presented in Table S1. Furthermore, to assess the portability of the daily scale prediction model and KF assimilation technique developed in this study, daily scale water quality monitoring data were collected from the Wofoxia (WFX) section in the upper reaches of the Dahei River, covering the period from May 2022 to August 2024, with the same parameters as those monitored in the RHK section.

2.2. The Xiaoqing River basin

To further explore the portability of the methods used in this study, we conducted research in the Xiaoqing River Basin (Fig. S1), a basin with significantly different climatic conditions from the Dahei River Basin. The Xiaoqing River Basin is located in Shandong Province, China. It flows from southwest to northeast and drains into Laizhou Bay in Weifang City. The Xiaoqing River Basin has obvious seasonal changes

Table 1
The statistical descriptions for all the daily monitoring data in Dahei river basin.

Parameters	Min	Max	Mean	Medians	SD	Skewness	Kurtosis
tavg (°C)	−26.800	28.400	7.536	8.900	12.480	−0.328	−1.090
tmin (°C)	−30.000	23.600	0.461	1.600	12.549	−0.290	−1.097
tmax (°C)	−17.000	36.000	15.361	16.900	12.558	−0.355	−1.064
prcp (mm)	0.000	103.100	1.250	0.000	5.865	9.671	125.875
snow (mm)	0.000	119.000	0.377	0.000	4.038	18.866	467.733
wdir (Degrees)	0.000	359.000	119.815	66.000	126.621	0.719	−1.021
wspd (km/h)	0.000	32.900	11.034	10.100	3.984	1.163	2.298
pres (hPa)	0.000	1045.300	1003.387	1017.600	120.805	−8.149	64.866
Flow (m^3/s)	0.350	21.400	2.888	2.700	1.345	5.058	42.962
Water temperature (°C)	0.400	30.200	13.191	12.550	8.920	0.116	−1.504
pH	6.000	9.000	8.005	8.000	0.413	0.220	2.815
DO (mg/L)	0.400	22.900	9.489	9.700	3.533	0.071	0.391
EC (S/m)	0.050	5283.500	1440.361	1362.100	590.559	0.876	2.047
TB (NTU)	0.200	2200.600	119.032	70.350	166.161	5.780	53.073
COD_{Mn} (mg/L)	0.700	23.200	7.358	7.100	3.729	0.682	0.200
NH_3N (mg/L)	0.020	14.040	0.860	0.240	1.636	4.030	20.459
TP (mg/L)	0.019	2.374	0.232	0.148	0.278	3.875	21.127
TN (mg/L)	0.220	17.970	4.742	3.660	3.039	1.077	0.876

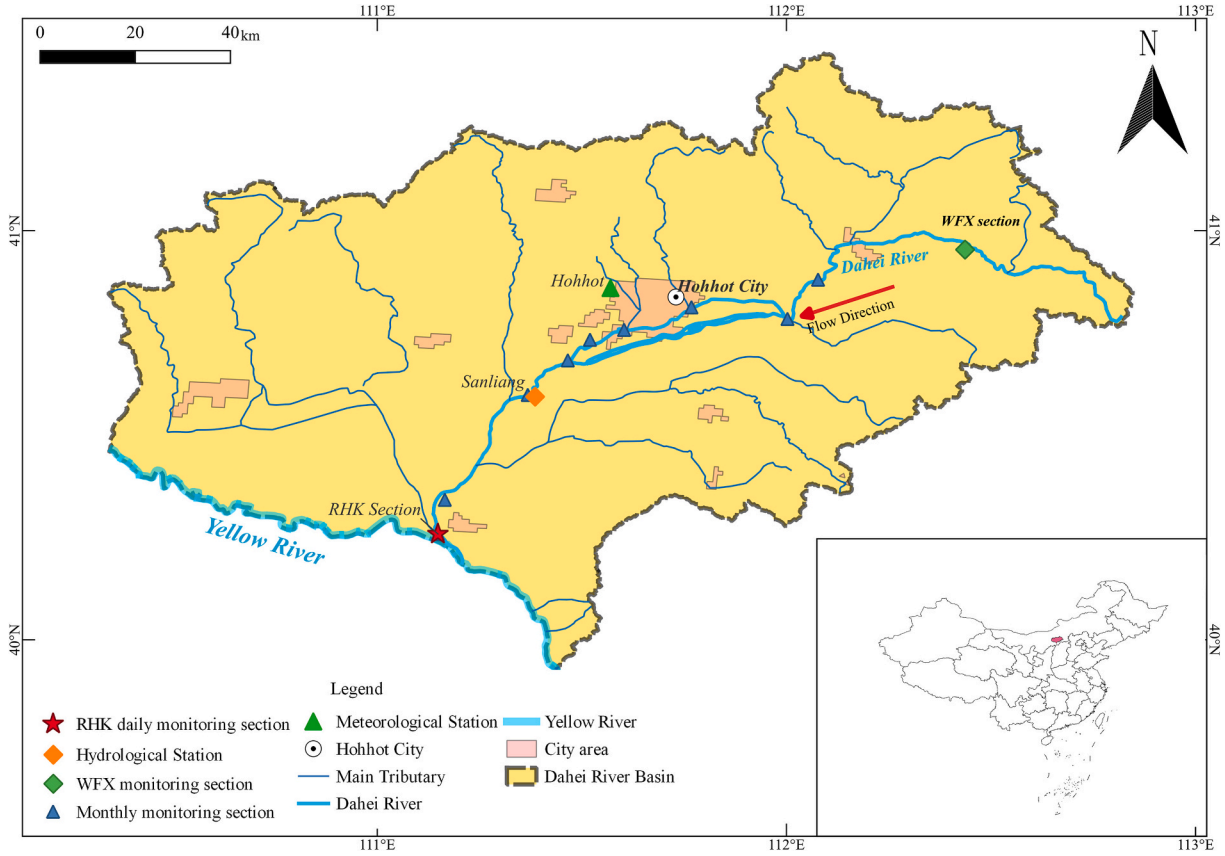


Fig. 1. Location of Dahei River Basin.

and belongs to the temperate monsoon climate type, with an average annual precipitation of approximately 620 mm. The basin is densely populated and industrially developed and includes a large area of agricultural cultivation. The Xiaoqing River has been seriously polluted for many years and is a key river under the control of local ecological and environmental departments. In this study, monitoring data from three sections (Mulizhuang (MLZ), Xinfengzhuang (XFZ) and Xizha (XZ)) of the Xiaoqing River mainstream from August 2018 to August 2021 were collected to study the model portability. The parameters of daily-scale data are the same as those of the Dahei River Basin.

3. Methodology

Fig. 2 presents an overview of the workflow and methodology implemented in this study. Details of this process are described in the following sections.

3.1. Kalman filters

The Kalman filter is a set of mathematical equations that provide a computationally efficient (recursive) method for estimating the state of a process in a manner that minimises the mean square error (Kalman, 1960). This filter is powerful in several ways: it enables the estimation of

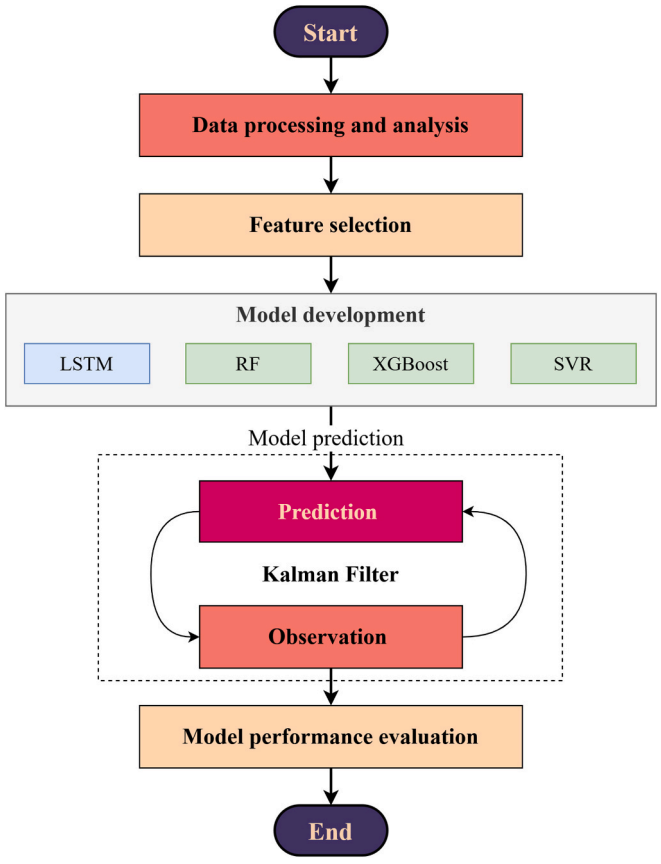


Fig. 2. Overview of the methodology used in the present study.

past, present, and even future states, and it can do so even when the precise nature of the modelled system is unknown.

The Kalman filter was the best-state estimator under linear and Gaussian noise assumptions. More precisely, if we denote the state as y_t and the measurement as z_t, we can assume the following model:

$$y_t = Ay_{t-1} + w, w \sim N(0, Q) \quad (-1)$$

$$z_t = Hy_t + v, v \sim N(0, R) \quad (-2)$$

where the matrix A represents the state at the previous time step $t-1$ to the state at the current step t; the matrix H represents the state to the measurement z_t; the matrices Q and R represent the process and measurement noise covariance, respectively, and the Kalman filter provides the best estimate of $\hat{y}_t$ based on the sum of the squared errors.

The Kalman filter achieves optimality through an iterative feedback loop comprising two update steps: the prediction step and the update step (Coskun et al., 2017). During the prediction step, we estimate the mean and covariance of the current state, independent of the current measurement:

$$\hat{y}_t^{'} = A\hat{y}_{t-1} \quad (-3)$$

$$\hat{P}_t^{'} = A\hat{P}_{t-1}A^T + Q \quad (-4)$$

During the update step, we can calculate the optimal Kalman gain K_t and use it with our observed measurements $\hat{z}_t$ to estimate the mean and covariance of y_t:

$$K_t = \hat{P}_t^{'}H^T\left(H\hat{P}_t^{'}H^T + R\right)^{-1} \quad (-5)$$

$$\hat{y}_t = \hat{y}_t^{'} + K_t\left(\hat{z}_t - H\hat{y}_t^{'}\right) \quad (-6)$$

$$\hat{P}_t = (I - K_tH)\hat{P}_t^{'} \quad (-7)$$

In this study, the Kalman filter was implemented using the Pykalman library in Python. Six parameters, namely 'transition matrices', 'observation matrices', 'initial state mean ', 'initial state covariance', 'observation covariance', and 'transition covariance', were preconfigured. The expectation–maximisation (EM) algorithm was employed to optimise these parameters iteratively. In this study, KF was mainly used to smooth the prediction results of several ML models to improve the prediction accuracy. For the four models, the specific KF hyperparameters are shown in Table 2.

3.2. LSTM

Unlike feedforward networks, recurrent neural networks (RNNs) are naturally suitable for modelling sequential data. However, variants such as simple RNNs are extremely difficult to train because of limitations such as vanishing gradient problems. LSTM neural networks solve these limitations by dynamically integrating gate functions and states into their architecture and have become one of the most widely used RNN architectures (Coskun et al., 2017). For an especial time stage t, the mathematical equations for the LSTM structure are defined as follows (Fang et al., 2021):

$$f_t = \sigma\left(W_{fx}X_t + W_{fh}h_{t-1} + b_f\right) \quad (8)$$

$$i_t = \sigma(W_{ix}X_t + W_{ih}h_{t-1} + b_i) \quad (9)$$

$$\widetilde{C}_t = \tanh(W_{cx}X_t + W_{ch}h_{t-1} + b_c) \quad (10)$$

$$C_t = f_t{*}C_{t-1} + i_t{*}\widetilde{C}_t \quad (11)$$

$$h_t = O_t{*}\tanh(C_t) \quad (12)$$

$$O_t = \sigma(W_{ox}X_t + W_{oh}h_{t-1} + b_o) \quad (13)$$

where X_t represents the input vector, while f_t, i_t, O_t are referred to the forget gate, input gate and output gate, respectively, and σ demotes the sigmoid function. W_{fx}, W_{ix}, W_{ox} represent the matrix of weights from the forget, input, and output gates to the input, respectively, W_{fh}, W_{ih}, and W_{oh} represent the matrix of weights from the three gates to the output, respectively, b_f, b_i, b_o are also the bias vectors of the three gates, respectively. $\widetilde{C}_t$ memorizes the up-to date data, C_t is the memory cell in time stage t, with C_{t-1} and h_{t-1} denoting the previous cell and its output vector, respectively, and h_t represents the output vector.

3.3. ML models

Random forest, an ensemble learning method, amalgamates multiple decision trees to formulate predictions (Breiman, 2001). Each decision tree undergoes independent training, facilitated by random sampling

Table 2
The initial hyperparameters of the Kalman filter with four ML models.

Model	Initial hyperparameters for KF
LSTM-KF	transition matrices = [1], observation matrices = [1], initial state mean = observation[0], initial state covariance = 1, observation covariance = 1, transition covariance = 1, kf.em (n_iter = 10)
RF-KF	transition matrices = [1], observation matrices = [1], initial state mean = observation[0], initial state covariance = 1, observation covariance = 1, transition covariance = 1, kf.em (n_iter = 10)
XGBoost-KF	transition matrices = [1], observation matrices = [1], initial state mean = observation[0], initial state covariance = 1, observation covariance = 0.01, transition covariance = 10, kf.em (n_iter = 20)
SVR-KF	transition matrices = [1], observation matrices = [1], initial state mean = observation[0], initial state covariance = 1, observation covariance = 100, transition covariance = 100, kf.em (n_iter = 50)

and feature selection of the input data, thereby enhancing the model's diversity (Wang et al., 2024). The final prediction of random forests is derived through a voting or averaging mechanism among the predictions of individual decision trees (Hou et al., 2024).

Random forests offer several advantages. Firstly, they exhibit robust performance when confronted with large datasets due to the ability to train each decision tree in parallel. Secondly, random forests demonstrate proficiency in handling high-dimensional data and features characterized by complex interrelationships. Moreover, they facilitate the evaluation of feature importance, thereby aiding researchers in discerning the pivotal features crucial for prediction tasks (Bae et al., 2018; Wu et al., 2019).

XGBoost stands as a formidable machine learning algorithm renowned for its efficacy in optimizing model predictions. Rooted in the gradient boosting framework, XGBoost operates as an optimization algorithm tailored to enhance prediction accuracy. Its core principle involves iteratively augmenting prediction outcomes by incorporating new models, each of which serves as a refinement of the residual error left by the preceding model. Employing a strategy termed "leaf-by-leaf splitting," XGBoost optimizes model performance by strategically splitting at each leaf node of the decision tree. This approach not only mitigates computational overhead but also accelerates training speed (Lu and Ma, 2020).

Moreover, XGBoost integrates advanced techniques such as parallel computing, cache optimization, and block compression to further augment algorithmic performance. These enhancements contribute significantly to refining the efficiency and scalability of the XGBoost algorithm, thereby reinforcing its status as a state-of-the-art tool in the realm of machine learning.

Support Vector Machine Regression (SVR) stands as a prevalent machine learning algorithm employed to address regression challenges (Vapnik, 2020). Rooted in the principle of Support Vector Machine (SVM), SVR orchestrates regression predictions by projecting data into a high-dimensional feature space and discerning an optimal hyperplane to approximate the genuine data distribution. SVR harnesses a kernel function technique to effectuate data mapping into a high-dimensional feature space, thereby enhancing its capacity to capture intricate nonlinear relationships within the data (Meyers et al., 2017).

SVR boasts several advantages over traditional regression models. Primarily, its adeptness in tackling nonlinear quandaries stems from its capability to leverage kernel functions for data projection into high-dimensional feature spaces. Moreover, SVR exhibits robust generalization prowess by relying solely on support vectors rather than all data points. Finally, SVR's efficiency in handling high-dimensional data is noteworthy, as it conducts computations seamlessly within high-dimensional feature spaces without succumbing to the curse of dimensionality.

3.4. Data preparation and model performance evaluation

Data preprocessing is mainly based on the pandas library in Python. For example, the pivot function is used to process the table data so that the format of the data set meets the experimental requirements. For outliers and missing values, we use linear interpolation or mean filling. The dataset was partitioned, with 70 % allocated for training and the remaining 30 % for testing. The training dataset facilitated the model development, whereas the test dataset was used for validation. To maintain uniformity across all the models, we applied minimum–maximum scaling techniques to normalise both the input and output variables. Furthermore, we transformed the dataset into a supervised learning framework, enabling the conversion of time-series data into input–output pairs.

In this study, we used the coefficient of multiple determination (R^2), the root mean squared error (*RMSE*), the mean absolute error (MAE), the mean absolute percentage error (MAPE) and the Nash–Sutcliffe Efficiency coefficient (NSE) to evaluate the performance of the water quality prediction models. The formulations were as follows:

$$R^2 = \frac{\left[\sum_{i=1}^{n}(y_i-\bar{y})(\hat{y}_i-\hat{\bar{y}}_i)\right]^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2\sum_{i=1}^{n}(\hat{y}_i-\hat{\bar{y}}_i)^2} \tag{14}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{n}} \tag{15}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i-y_i| \tag{16}$$

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i-\hat{y}_i}{y_i}\right| \tag{17}$$

$$NSE = 1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2} \tag{18}$$

where, y_i represents the ith observation value; $\hat{y}_i$ represents the ith prediction value; $\bar{y}$ represents the average of water quality observation; $\hat{\bar{y}}_i$ represents the mean water quality value from the prediction model; and n is the number of observed samples.

3.5. Water quality assessment and trend analysis

The water quality index (WQI) methodology has been extensively employed in the assessment of surface and groundwater quality (Pesce and Wunderlin, 2000). This approach surpasses the traditional single-factor evaluation methods by integrating multiple physical and chemical indicators. Consequently, it provides a more comprehensive and holistic reflection of the current water quality status. The pollution-sharing ratio serves as an analytical tool for assessing the influence of diverse pollution sources on the water quality within aquatic systems. This method facilitates the identification and analysis of pollution sources and their underlying causes by quantifying the proportional contribution of individual pollution indicators to the overall water quality.

Mann–Kendall (MK) trend analysis is a statistical approach extensively utilised for discerning trends and their significance within time-series data (Gocic and Trajkovic, 2013). Using the Mann–Kendall test, this method ascertains whether a time series exhibits a monotonic trend over a specified period. Notably, this analytical technique operates independently of any presumptions regarding the distribution of the data, rendering it applicable for scrutinising datasets that may not adhere to a normal distribution. Because of its robustness and versatility, the Mann–Kendall method has found widespread application across various domains, including environmental and climatological studies. In this study, we employed the WQI method along with the pollution sharing rate to assess the water quality monitoring data collected from the RHK section. Additionally, we utilised the MK method to scrutinize the multi-year trends of each water quality parameter. More details about these methods can be found in this literature (Gao et al., 2023). These methodologies were utilised to identify the principal pollution indicators, which were subsequently designated as the target features for the prediction models.

4. Results

4.1. Water quality assessment of RHK section

The water quality assessment outcomes revealed a WQI of 1.77 for the RHK section, indicating a state of heavy pollution. Nevertheless, the results of the MK trend analysis revealed a discernible trend towards a gradual reduction in water quality pollution levels. The analysis of pollution sharing rates further delineated TN as the primary pollution

indicator, accounting for the highest proportion (55.3 %), followed by COD_{Mn} (16.6 %) and TP (11.7 %). Based on the criteria for surface water Category III, the multiple of TN exceedance was 4.74, while those for TP and COD_{Mn} were 1.16 and 1.23, respectively. Given the fundamental characteristics and land-use patterns within the Dahei River Basin, the downstream region predominantly serves as a vast agricultural cultivation and irrigation zone. The RHK section is located at the end of the Dahei River and directly reflects the influence of the Dahei River on the water quality of the Yellow River. Hence, the selection of TN, TP, and COD_{Mn} as the key water quality parameters for prediction aligns with the priorities of local authorities, given their significant contribution to pollution and corresponding regulatory concerns.

4.2. Predictive model implementation

The LSTM prediction model architecture includes an input layer, hidden layer, dropout layer, and fully connected layer (Table 3). The input layer integrates nine features and time steps with a rectified linear unit (ReLU) activation function employed in the hidden layer. To mitigate overfitting, a dropout layer was incorporated at a rate of 0.001 after the LSTM layer. We utilised a fully connected layer named 'dense' with a single unit and a linear activation function to finalise the model configuration. Subsequently, the model was compiled using the MAE loss function and adaptive moment estimation (Adam) optimiser with a learning rate of 0.01. Throughout the model training phase, hyperparameters, including epochs and batch sizes, were tailored to accommodate variations across different prediction indicators. To evaluate the model's performance and ensure an optimal fit, we plotted a learning curve during the experiment to discern the indications of overfitting or underfitting.

During the construction of the RF model, a tenfold cross-validation grid search method was employed to fine-tune the parameters (Table 3) with the random_state parameter set to 42. For predicting TN, the hyperparameters were configured with a maximum depth (max_depth) of five, the number of estimators (n_estimators) was 100, and the minimum samples leaf (min_samples_leaf) was 1. In the prediction of TP, the max_depth parameter remained at five, whereas the number of estimators increased to 300 and the minimum samples leaf was 5. Regarding the prediction of COD_{Mn}, the maximum depth parameter was left unrestricted (none), the number of estimators was set to 100, and the minimum samples leaf was 5.

The primary optimised parameters of the XGBoost model formulated in this investigation included n_estimators, learning rate, max_depth, min_child_weight, gamma, subsample and colsample_bytree (Table 2). In addition, employing a cross-validation methodology facilitates the fine-tuning of these parameters. In the context of TN prediction, the optimal values identified were 100, 0.1, 5, 1, 0, 0.8 and 0.8, respectively; for TP prediction, the preferred parameter configuration consisted of 200, 0.01, 5, 1, 0.1, 0.8 and 1.0. Similarly, for COD_{Mn} prediction, the most effective parameter values were 100, 0.01, 7, 1, 0, 0.8 and 0.8.

An SVR model was constructed using the epsilon–SVR kernel equations. During the development of the SVR model, various kernel functions were explored, including linear and radial basis functions (RBF), sigmoids, and polynomials. Other hyperparameters for training the SVR model included the adjustment factor (C), epsilon (ε), gamma (γ), degree and coef0 (Table 3). A tenfold cross-validation grid search method was employed to determine the optimal hyperparameters iteratively. For the TN prediction model, the kernel is 'poly', the optimal values of C, ε, degree, and coef0 were determined as 0.7, 0.0001, 5, and 5, respectively. Similarly, the optimal values for the TP prediction model were 'poly', 0.7, 0.001, 2, and 5, whereas those for the COD_{Mn} prediction were 'poly', 0.7, 0.0001, 4, and 4.

Table 3
The structures and hyperparameters of the developed water quality parameters prediction models.

Model	Predictive variable	Structure and hyperparameters
LSTM	TN	1 LSTM layer (100 neurons, ReLU activation) + 1 Dropout layer (0.001) + 1 Dense layer (1 neuron, linear activation) + Compile (MAE loss, Adam optimizer with a learning rate of 0.01)
	TP	1 LSTM layer (200 neurons, ReLU activation) + 1 Dropout layer (0.001) + 1 Dense layer (1 neuron, linear activation) + Compile (MAE loss, Adam optimizer with a learning rate of 0.01)
	COD_{Mn}	1 LSTM layer (200 neurons, ReLU activation) + 1 Dropout layer (0.001) + 1 Dense layer (1 neuron, linear activation) + Compile (MAE loss, Adam optimizer with a learning rate of 0.01)
RF	TN	maximum depth (max_depth) = 5, the number of estimators (n_estimators) = 100, max_features = 0.6, min_samples_split = 2, min_samples_leaf = 1
	TP	maximum depth (max_depth) = 5, the number of estimators (n_estimators) = 300, max_features = 0.6, min_samples_split = 2, min_samples_leaf = 5
	COD_{Mn}	maximum depth (max_depth) = None, the number of estimators (n_estimators) = 100, max_features = 0.6, min_samples_split = 2, min_samples_leaf = 5
XGBoost	TN	n_estimators = 100, learning_rate = 0.1, max_depth = 5, min_child_weight = 1, gamma = 0, subsample = 0.8, colsample_bytree = 0.8
	TP	n_estimators = 200, learning_rate = 0.01, max_depth = 5, min_child_weight = 1, gamma = 0.1, subsample = 0.8, colsample_bytree = 1.0
	COD_{Mn}	n_estimators = 100, learning_rate = 0.01, max_depth = 7, min_child_weight = 1, gamma = 0, subsample = 0.8, colsample_bytree = 0.8
SVR	TN	kernel = 'poly', C = 0.7, epsilon = 0.0001, degree = 5, coef0 = 5
	TP	kernel = 'poly', C = 0.7, epsilon = 0.001, degree = 2, coef0 = 5
	COD_{Mn}	kernel = 'poly', C = 0.7, epsilon = 0.0001, degree = 4, coef0 = 4

4.3. Performance of water quality prediction model

One of the primary objectives of this research was to assess the predictive capabilities of four ML models (LSTM, RF, XGBoost, and SVR) for forecasting three water quality parameters. The models were trained using the provided training dataset, after which the testing dataset was employed to predict the target features. The detailed results are presented in Table 4. Evidently, LSTM had the highest predictive performance during the training and testing periods, followed by RF, with XGBoost and SVR having slightly worse predictive accuracies.

LSTM consistently achieved the best predictive performance for TN, with the highest R^2 (0.885), the lowest RMSE (0.864 mg/L), MAE (0.490 mg/L), and MAPE (10.93 %) during the testing period. Additionally, its NSE value (0.877) was the highest among all models, indicating strong agreement between predictions and observations. RF and SVR showed moderate performance, while XGBoost exhibited the weakest performance for TN prediction, with the lowest R^2 (0.778) and highest errors. For TP prediction, LSTM again outperformed the other models during the testing period, with a high R^2 (0.871), low RMSE (0.034 mg/L), MAE (0.020 mg/L), and a relatively low MAPE (13.83 %). RF also achieved strong results (R^2 = 0.848, NSE = 0.849). In contrast, XGBoost and SVR performed poorly on TP, particularly SVR, which yielded the lowest R^2 (0.555) and the highest MAPE (33.78 %), suggesting limitations in capturing the TP dynamics. Although all models demonstrated lower predictive accuracy for COD_{Mn} compared to TN and TP, LSTM still showed superior generalization ability. It achieved the lowest RMSE (1.451 mg/L), MAE (0.824 mg/L), and MAPE (10.02 %) during testing, alongside an NSE of 0.839. RF followed closely, while XGBoost and SVR yielded comparatively poorer performance with higher prediction errors and lower NSE values.

A comparison of the model's predicted values for the three water quality parameters against the observed values provided an intuitive

Table 4
The results of R^2 and RMSE for the LSTM, RF, XGBoost and SVR model in training period and testing period in Dahei River RHK section.

Water quality parameter	Metrics	Training period				Testing period			
		LSTM	RF	XGBoost	SVR	LSTM	RF	XGBoost	SVR
	R^2	0.911	0.873	0.811	0.824	0.885	0.857	0.778	0.798
	RMSE (mg/L)	0.849	0.911	1.133	1.006	0.864	0.962	1.210	1.143
	MAE (mg/L)	0.467	0.562	0.788	0.661	0.490	0.595	0.803	0.676
	MAPE (%)	9.81	14.57	15.29	12.56	10.93	15.30	17.53	14.62
TN	*NSE*	0.897	0.872	0.799	0.832	0.877	0.858	0.769	0.798
	R^2	0.892	0.921	0.756	0.604	0.871	0.848	0.736	0.555
	RMSE (mg/L)	0.031	0.089	0.048	0.058	0.034	0.037	0.050	0.065
	MAE (mg/L)	0.018	0.022	0.030	0.038	0.020	0.024	0.036	0.041
	MAPE (%)	11.98	16.05	27.34	29.92	13.83	17.87	31.59	33.78
TP	*NSE*	0.892	0.887	0.754	0.765	0.871	0.849	0.725	0.701
	R^2	0.851	0.866	0.767	0.736	0.839	0.802	0.739	0.704
	RMSE (mg/L)	1.213	1.376	1.721	1.814	1.451	1.606	1.842	1.997
	MAE (mg/L)	0.809	0.946	1.112	1.298	0.824	0.970	1.186	1.496
	MAPE (%)	9.87	11.89	13.59	18.98	10.02	12.54	15.32	22.94
COD_{Mn}	*NSE*	0.857	0.823	0.789	0.750	0.839	0.802	0.739	0.697

assessment of the predictive efficacy of the four models (Figs. 3–5). Notably, the LSTM model demonstrated superior performance compared to the other models, which is consistent with the findings presented in Table 3. It is evident that the LSTM model outperformed the other three models in predicting TN, TP, and COD_{Mn}, particularly in accurately capturing the peaks and valleys. The SVR and XGBoost models exhibited considerable discrepancies between their predictions of extreme and observed values, and RF appeared to underestimate certain points. The RF model exhibited a good prediction performance compared to the other two models. While XGBoost demonstrates better performance in predicting TP and COD_{Mn} compared to the SVR model, its accuracy in predicting TN is inferior to that of the SVR model.

The box plots (Fig. 6) illustrate that the median of the predicted values from the LSTM model was closely aligned with the observed values for the three water quality indicators, whereas the median values of the other three models tended to be higher than the observed values. With respect to the data range (maximum and minimum values), the values predicted by the LSTM model were closest to the observed values. The predictions of the RF model for the TP and COD_{Mn} closely resembled the observed values. However, significant disparities are observed between the predictions of XGBoost and SVR, particularly for SVR. Regarding the quartiles, in the prediction of TN, the RF model closely approximated the observed values, while some discrepancy was observed between the third quartile of the LSTM model and the observed values. The disparities between SVR and XGBoost and the observed values were more pronounced. For predicting TP and COD_{Mn}, the LSTM model aligned closely with the observed values, whereas the quartile distribution range of the RF model was narrower than that of the observed values. Moreover, the distributions of the XGBoost and SVR exhibited significant deviations from the observed values. Overall, the prediction results of the LSTM model closely resembled the distribution of the observed values, whereas the predictions of the other three models displayed disparities in statistical variables compared with the observed values, indicating a lower prediction accuracy relative to the LSTM model.

4.4. Kalman filter improves model prediction accuracy

We conducted data assimilation of the four models using a KF. Subsequently, the assimilated models, LSTM–KF, RF–KF, XGBoost–KF, and SVR–KF, were employed to generate new predictions for the three water quality parameters. Table 5 presents a comparative analysis of the model prediction evaluation results post-assimilation, along with previous findings. Furthermore, to offer a more intuitive demonstration of the enhancement in prediction accuracy achieved through KF, multiple scatter plots were generated for comparative analysis (Fig. 7). This study also employed the percentage change in the RMSE post-assimilation to quantify the enhancement in the prediction performance achieved by KF across the four prediction models (Fig. 8).

For TN prediction, KF significantly enhanced the prediction accuracy of all four models. In particular, the LSTM-KF model achieved the best overall results, with R^2 increasing from 0.885 to 0.909, and NSE rising from 0.877 to 0.901. Error metrics also showed substantial reductions: RMSE decreased from 0.864 mg/L to 0.768 mg/L (an 11.1 % reduction), MAE dropped from 0.490 mg/L to 0.440 mg/L (10.2 % decrease), and MAPE decreased by 12.3 %, from 10.93 % to 9.59 %. Similar trends were

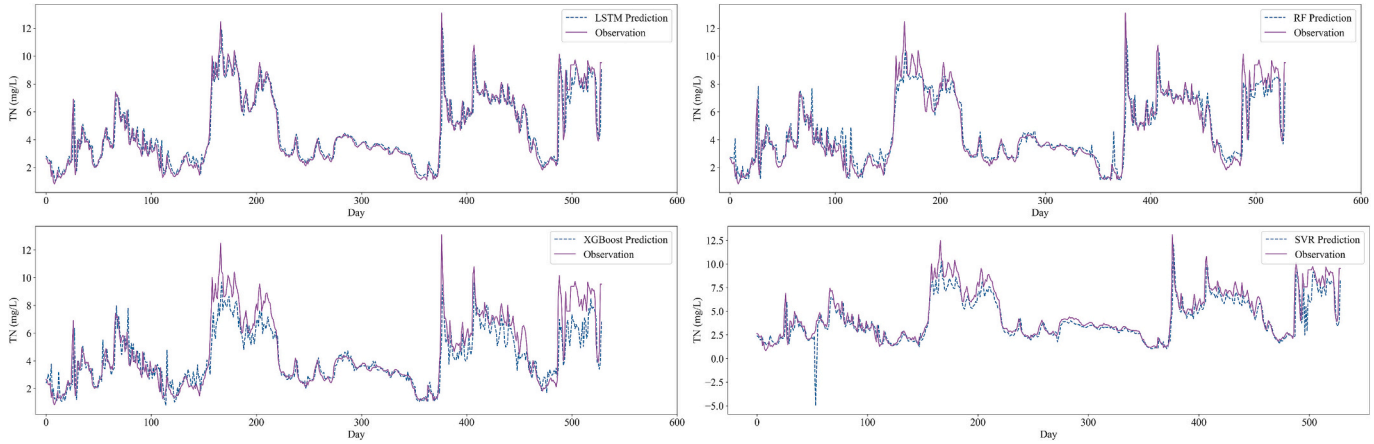


Fig. 3. The curves of prediction and observation for TN based on four models in the RHK section.

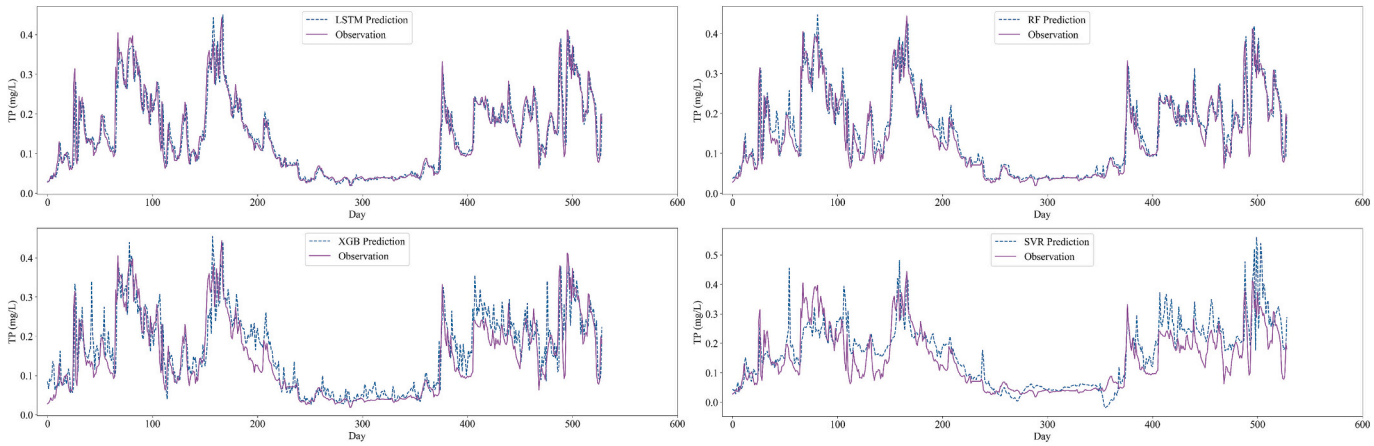


Fig. 4. The curves of prediction and observation for TP based on four models in the RHK section.

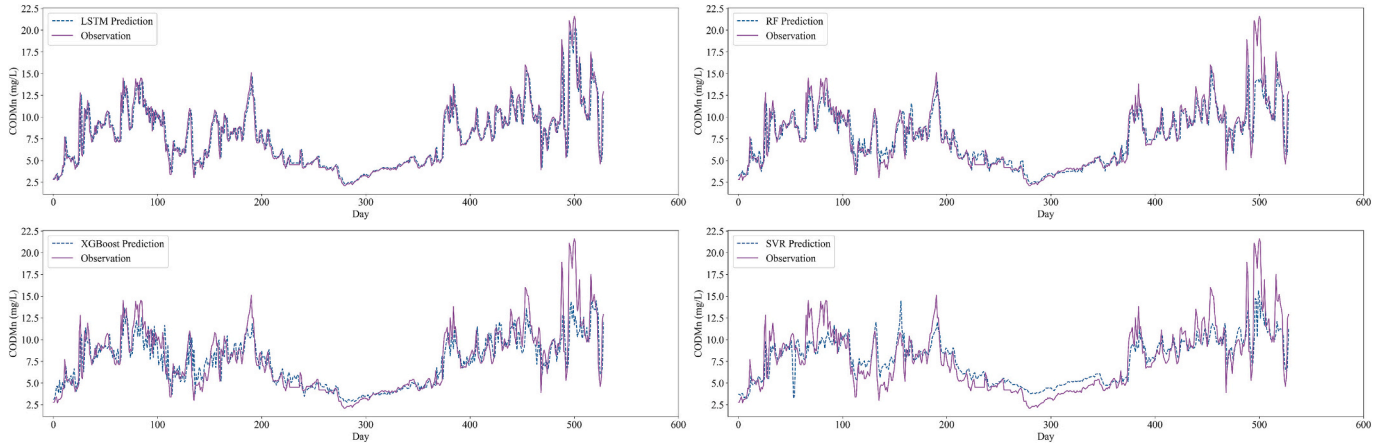


Fig. 5. The curves of prediction and observation for COD_{Mn} based on four models in the RHK section.

observed for other models. For example, the RF-KF model reduced its RMSE by 10.8 % and MAPE by 11.9 % compared to RF, while SVR-KF improved its NSE from 0.798 to 0.840 and reduced its MAPE by 10.4 %. In the case of TP prediction, the integration of KF yielded particularly prominent gains in accuracy. The LSTM-KF model improved R^2 from 0.871 to 0.910 and NSE from 0.871 to 0.907. RMSE and MAE were reduced by 17.6 % and 15.0 %, respectively. Similarly, RF-KF showed a 16.2 % reduction in RMSE and a 12.5 % drop in MAE. The XGBoost-KF model exhibited a 16.0 % improvement in RMSE and 11.1 % in MAE. Despite SVR initially performing poorly on TP prediction, KF assimilation led to meaningful improvements, reducing RMSE by 9.2 %, MAE by 7.3 %, and increasing NSE from 0.701 to 0.754.

For COD_{Mn} prediction, all models benefited from KF assimilation, albeit with slightly smaller margins compared to TN and TP. The LSTM-KF model improved R^2 from 0.839 to 0.876 and NSE from 0.839 to 0.875. Its RMSE decreased from 1.451 mg/L to 1.275 mg/L (12.1 % reduction), while MAPE dropped by 10.8 %. The RF-KF model reduced its RMSE and MAE by 10.7 % and 8.7 %, respectively. Although SVR remained the least accurate model, SVR-KF still achieved minor improvements in RMSE (from 1.997 to 1.911 mg/L) and MAPE (from 22.94 % to 22.81 %).

4.5. ML–KF method transplantation test

In this study, a prediction model data assimilation method was applied to other daily monitoring sections and monthly monitoring datasets. Specifically, an LSTM–KF prediction model was constructed for the WFX section to predict three water quality indicators: TN, TP, and COD_{Mn}. The results demonstrated that this method effectively improved prediction accuracy. The R^2 values for these three indicators increased from 0.805, 0.601, and 0.650 to 0.860, 0.684, and 0.746, respectively. Simultaneously, the RMSE values decreased from 0.464, 0.009, and 0.755 mg/L to 0.395, 0.006, and 0.643 mg/L, respectively. Scatter plots before and after assimilation (Fig. S2) reveal that, as with the RHK section, the post-assimilation results were closer to the 1:1 line, indicating enhanced prediction accuracy.

Currently, most water quality monitoring sections in the study area are monitored monthly because full coverage by daily automated monitoring stations has not yet been achieved. Therefore, we developed a prediction model based on monthly monitoring data from the Dahei and Xiaohei rivers, incorporating KF technology to enhance the prediction accuracy and better meet the needs of water environment managers. Given the long-time intervals between the monthly monitoring data, which significantly reduced temporal correlations, the RF algorithm was selected to build the prediction model. The prediction results of the RF model for the three parameters, as well as the enhanced RF–KF results, are listed in Table S2. These findings demonstrate that the proposed method improves the accuracy of monthly prediction models and exhibits a strong predictive performance.

To further validate the generalization ability of the proposed LSTM-KF model, we conducted a transferability analysis in the Xiaoqing River Basin. The results of water quality prediction for TN, TP, and COD_{Mn} based on LSTM and LSTM-KF models are summarized in Table 6. Across all stations and indicators, the LSTM-KF model consistently

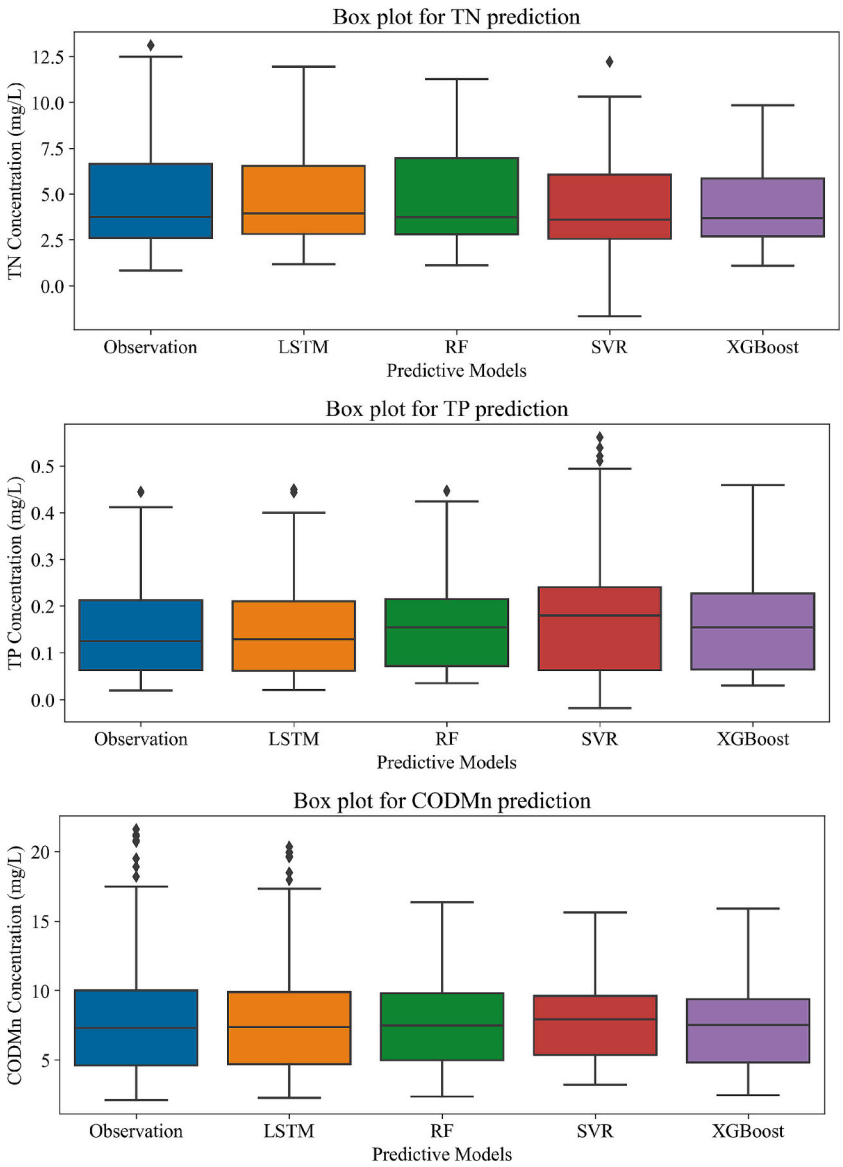


Fig. 6. Box plots comparing the predictive model's abilities for TN, TP and COD_{Mn}.

Table 5
Evaluation of Predictive Performance after Incorporating Kalman Filtering Algorithm into Prediction Models.

Water quality parameter	Metrics	LSTM	LSTM-KF	RF	RF-KF	XGBoost	XGBoost-KF	SVR	SVR-KF
	R^2	0.885	0.909	0.857	0.886	0.778	0.797	0.798	0.840
	RMSE (mg/L)	0.864	0.768	0.962	0.858	1.210	1.132	1.143	1.016
	MAE (mg/L)	0.490	0.440	0.595	0.540	0.803	0.736	0.676	0.629
	MAPE (%)	10.93	9.59	15.30	13.48	17.53	15.18	14.62	13.10
TN	*NSE*	0.877	0.901	0.858	0.886	0.769	0.797	0.798	0.840
	R^2	0.871	0.910	0.848	0.896	0.736	0.813	0.555	0.625
	RMSE (mg/L)	0.034	0.028	0.037	0.031	0.050	0.042	0.065	0.059
	MAE (mg/L)	0.020	0.017	0.024	0.021	0.036	0.032	0.041	0.038
	MAPE (%)	13.83	12.43	17.87	16.53	31.59	28.93	33.78	30.03
TP	*NSE*	0.871	0.907	0.849	0.896	0.725	0.804	0.701	0.754
	R^2	0.839	0.876	0.802	0.842	0.739	0.776	0.704	0.719
	RMSE (mg/L)	1.451	1.275	1.606	1.434	1.842	1.691	1.997	1.911
	MAE (mg/L)	0.824	0.735	0.970	0.886	1.186	1.091	1.496	1.474
	MAPE (%)	10.02	8.94	12.54	12.12	15.32	14.06	22.94	22.81
COD_{Mn}	*NSE*	0.839	0.875	0.802	0.842	0.739	0.775	0.697	0.711

outperformed the conventional LSTM model. For TN, the introduction of KF led to notable improvements in prediction accuracy. At MLZ, RMSE decreased from 0.262 mg/L to 0.224 mg/L and R^2 increased from 0.804 to 0.856. Similar performance gains were observed at XFZ and XZ, with R^2 values improving from 0.889 to 0.915 and from 0.845 to 0.891, respectively. Corresponding reductions in MAPE and enhancements in

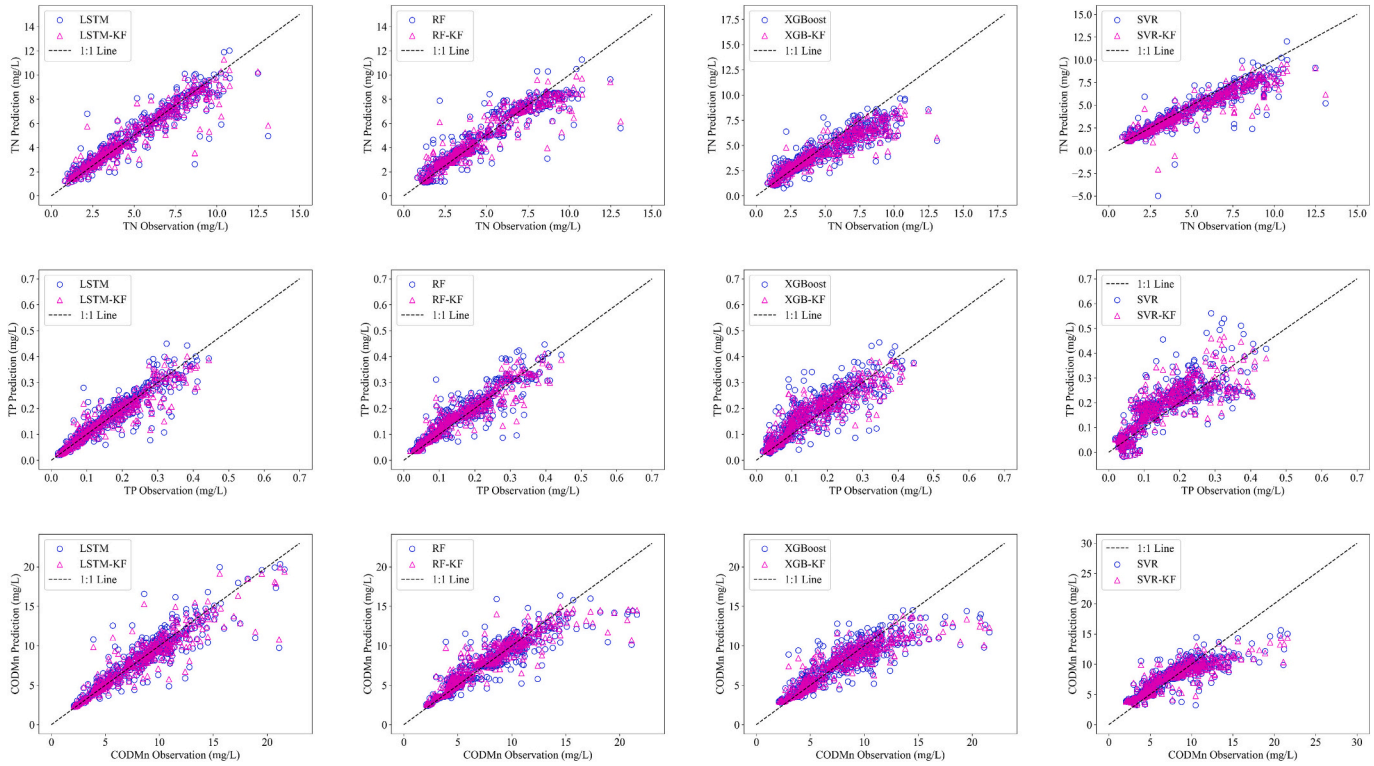


Fig. 7. Scatter plots of predicted and observed values, with Kalman-filtered assimilated data.

NSE also confirmed the effectiveness of the KF component in reducing model uncertainty. For TP prediction, LSTM-KF achieved overall higher accuracy at all three sites. At MLZ, the RMSE dropped from 0.013 to 0.011 mg/L, and NSE improved from 0.648 to 0.756. At XFZ and XZ, the MAPE decreased by over 1 % compared to LSTM, and NSE values were also consistently improved (0.801 and 0.770, respectively). The prediction of COD_{Mn} also benefited from Kalman filtering. At MLZ, RMSE declined from 0.561 to 0.460 mg/L, while R^2 rose from 0.687 to 0.789. At XFZ and XZ, LSTM-KF achieved lower prediction errors and higher NSE values compared to LSTM, with improvements in MAPE ranging from 0.7 % to 1.9 %. These results demonstrate that the LSTM-KF model can be effectively transferred to different basins and maintain its predictive accuracy across varying hydrological and pollution conditions.

These quantitative improvements were further corroborated by the scatter plots of predicted versus observed values (Fig. 9), which visually illustrate the enhancement brought by Kalman filtering. For all three water quality indicators at the three monitoring sections, the scatter points of the LSTM-KF model were more tightly clustered around the 1:1 reference line compared to those of the standalone LSTM model. This pattern indicates reduced prediction bias and variance, further confirming the robustness and reliability of the LSTM-KF model in different hydrological contexts.

5. Discussion

5.1. LSTM vs. traditional ML models

Based on the findings presented in Figs. 3–6 and Table 4, the overall ranking of the water quality parameter prediction accuracy in the RHK section is LSTM > RF > XGBoost > SVR. The LSTM model exhibited the highest accuracy, with an average R^2 of 0.865. The RF model also performed well, achieving an average R^2 of 0.836; however, it had a larger RMSE than the predictions of the LSTM model'. Conversely, both XGBoost and SVR demonstrate higher prediction errors, with SVR exhibiting the poorest performance. These observations are consistent with the conclusions of previous studies (Barzegar et al., 2020; Bui et al., 2020; Huang et al., 2022). The superior performance of LSTM is primarily attributable to its inherent strength in modelling nonlinear and long-range temporal dependencies in time-series data. Unlike traditional machine learning models such as RF, XGBoost, and SVR—which, while proficient at learning nonlinear patterns, lack explicit mechanisms to account for temporal correlations—the LSTM architecture incorporates memory cells and gating mechanisms that allow it to retain and utilize historical information over extended sequences. This characteristic makes LSTM particularly effective in capturing delayed or cumulative effects in water quality evolution. This is consistent with the conclusions of other related studies (Huang et al., 2024; Prasad et al., 2022; Than et al., 2021). This advantage was particularly evident in the prediction of parameters such as TP and COD_{Mn}, where temporal fluctuations are more pronounced. LSTM consistently achieved lower MAE and MAPE values, while also maintaining higher NSE scores compared to its machine learning counterparts. Given the time-dependent nature of water quality monitoring data, deep learning models like LSTM present a distinct advantage in accurately modelling the complex dynamics of aquatic systems. Their capacity to learn both the magnitude and temporal trend of water quality variations justifies their selection for predictive tasks in this study, and they are expected to deliver higher accuracy than traditional ML models.

5.2. KF assimilation ML model significantly improves prediction accuracy

According to the scatter plot in Fig. 7, it can be seen that KF could effectively improve the prediction accuracy of the four prediction models. For the dominant pollutant TN, the accuracy hierarchy became LSTM-KF > RF-KF > SVR-KF > XGBoost-KF; for TP and COD_{Mn} the order remained LSTM-KF > RF-KF, with XGBoost-KF and SVR-KF alternating in the lower positions. The integration of the model with the KF does not alter the original structure or operational logic of the model. Therefore, the results continue to highlight the advantages of the LSTM model in processing time-series data, attributed to its structural characteristics

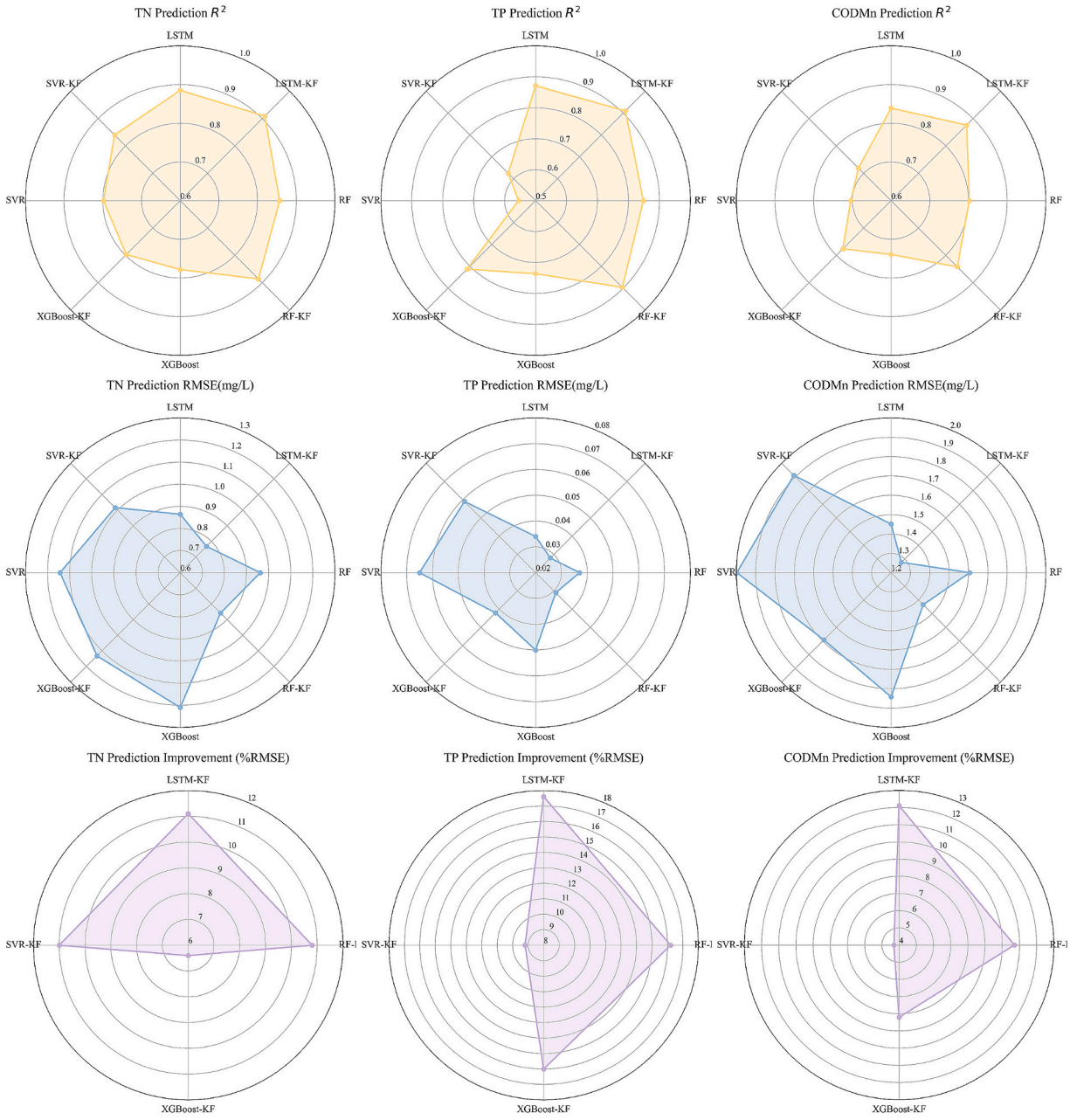


Fig. 8. Comparison of performance evaluation before and after assimilation with Kalman filtering in prediction models.

and exceptional memory capability. Our findings are consistent with previous studies. For instance, Chen et al. (2021b) integrated the LSTM model with TrAdaBoost for dissolved oxygen prediction, achieving a 15 %–25 % improvement in accuracy. Similarly, Zhang et al. (2022) proposed an EMD-LSTM water quality prediction model, which demonstrated an R^2 increase of 1.6 %–2.8 % compared to traditional models.

Across the three water-quality variables—TN, TP and COD_{Mn}—KF increased the mean R^2 of the four baseline models from 0.827 to 0.867, while simultaneously reducing RMSE, MAE and MAPE by 10–18 % on average. The gains were most prominent for the sequential LSTM model (Bakhshi Ostadkalayeh et al., 2023; Dong et al., 2023; Tian et al., 2024; Ye et al., 2024), yet even the non-sequential RF, XGBoost and SVR benefitted markedly, underscoring the general utility of KF as a lightweight post-processing step.

For TN, KF raised the R^2 of LSTM from 0.885 to 0.909 and lowered its RMSE by 11.1 % (0.864 → 0.768 mg/L). Similar improvements were observed for RF (RMSE −10.8 %, MAPE −11.9 %) and SVR (RMSE −11.1 %, NSE +0.042). These reductions imply KF's ability to dampen high-frequency noise and correct short-term biases that tree-based and kernel models do not explicitly address (Chen et al., 2022; Liu et al., 2022; Luo et al., 2023).

Improvements were even more striking for TP, whose low concentrations and episodic peaks pose particular challenges. LSTM-KF achieved a 17.6 % drop in RMSE (0.034 → 0.028 mg/L) and an NSE increase from 0.871 to 0.907. RF-KF and XGBoost-KF followed the same pattern, reducing their RMSE by 16.2 % and 16.0 %, respectively, and pushing NSE values above 0.80. These results show that assimilating real-time observations mitigates the tendency of non-linear regressors to over- or under-estimate extreme TP events.

Although COD_{Mn} remained the most difficult variable, KF still yielded measurable benefits (mean RMSE −12 %; mean NSE +0.036). For instance, LSTM-KF lowered RMSE from 1.451 to 1.275 mg/L and decreased MAPE from 10.02 % to 8.94 %. The prediction accuracy was also consistent with previous studies, and some RMSEs were smaller

Table 6
Evaluation of Predictive Performance after Incorporating Kalman Filtering Algorithm into LSTM Models in Xiaoqing River.

Water quality parameter	Metrics	MLZ		XFZ		XZ	
		LSTM	LSTM-KF	LSTM	LSTM-KF	LSTM	LSTM-KF
TN	R^2	0.804	0.856	0.889	0.915	0.845	0.891
	RMSE (mg/L)	0.262	0.224	0.542	0.474	0.665	0.557
	MAE (mg/L)	0.173	0.154	0.391	0.339	0.462	0.392
	MAPE (%)	18.84	17.58	6.27	5.45	7.15	6.20
	NSE	0.796	0.847	0.889	0.915	0.844	0.893
TP	R^2	0.645	0.757	0.721	0.795	0.663	0.766
	RMSE (mg/L)	0.013	0.011	0.028	0.024	0.031	0.025
	MAE (mg/L)	0.006	0.005	0.020	0.018	0.017	0.015
	MAPE (%)	14.79	13.57	11.43	10.19	12.34	10.85
	NSE	0.648	0.756	0.722	0.801	0.667	0.770
COD_{Mn}	R^2	0.687	0.789	0.628	0.741	0.670	0.763
	RMSE (mg/L)	0.561	0.460	0.696	0.580	0.623	0.541
	MAE (mg/L)	0.351	0.302	0.368	0.326	0.528	0.472
	MAPE (%)	22.60	20.86	8.36	7.63	9.59	8.71
	NSE	0.674	0.776	0.730	0.812	0.691	0.788

(van der Westhuizen et al., 2023; Xiong et al., 2023). This underscores its utility as a reference tool for predicting daily water quality changes.

Mechanistically, the Kalman update step fuses model predictions with incoming observations, weighting them by their respective error covariances. In our setting—dense, near-real-time monitoring—this fusion reduces propagation of past errors and prevents drift, thereby producing trajectories that adhere more closely to the 1:1 line (Fig. 7). The sharper clustering of LSTM-KF scatter points confirms reduced bias and variance, while the parallel error-metric improvements quantify the same effect. Collectively, these findings highlight that KF assimilation is a low-cost yet powerful means of enhancing both deep-learning and conventional machine-learning water-quality models, especially where continuous monitoring data are available for iterative correction.

5.3. Applicability of the method to different sections and time resolutions

According to Fig. S2 and Table S2, the ML–KF method proposed in this study is applicable to both the daily scale dataset of the WFX section and the monthly water quality monitoring dataset from the mainstream Dahei and Xiaohei rivers. These results demonstrate that the applicability of the proposed method is not restricted to specific monitoring sections or time resolutions. This method provides data and technical support to local ecological and environmental management personnel. For time-series monitoring data on a daily scale, a more accurate LSTM model can be employed, which, when combined with the KF method, achieves optimal prediction accuracy. For monthly scale monitoring data, in which the dataset for an individual section may be insufficient to support robust modelling, the approach of integrating data from multiple sections, as adopted in this study, can be utilised. This integration, when paired with data assimilation techniques, improves prediction accuracy and ensures the ‘utility of the method across diverse temporal and spatial scales.

In addition to the applicability of the method in the same basin, a test in the Xiaoqing River Basin showed that the method has good universality and portability. As shown in Table 6, both models demonstrated good adaptability across regions, with LSTM–KF consistently outperforming LSTM in terms of predictive accuracy for all three water quality parameters at each section (Gao et al., 2023; Luo et al., 2023; Wan et al., 2022).

Specifically, for TN, the R^2 values of the LSTM–KF model across the three sections ranged from 0.856 to 0.915, and RMSEs remained low (0.224–0.557 mg/L), indicating a high level of accuracy and stability. For TP, although the absolute concentrations were lower, the assimilation-enhanced model still achieved notable improvements, with R^2 values increasing by approximately 0.08–0.11 compared to the LSTM model. A similar trend was observed in COD_{Mn} predictions, where LSTM–KF improved R^2 by over 0.07 at each site, while reducing RMSE and MAE simultaneously.

These results suggest that the LSTM–KF model not only performs well within the original study area but also maintains strong predictive capabilities when applied to other watersheds with different hydrological and anthropogenic characteristics. This generalization ability likely stems from the LSTM’s inherent advantage in capturing temporal patterns and the KF’s capacity to correct systematic errors via real-time data assimilation. In practical applications, such cross-basin transferability is crucial, particularly when historical data from target regions are scarce. Therefore, the LSTM–KF framework offers a promising solution for regional water quality management, especially in data-limited or newly monitored basins.

5.4. Uncertainty analysis and future directions of the research

The results of this study and the evaluation confirm the effectiveness of the Kalman filter data assimilation method in improving the prediction accuracy of water quality parameters using a ML model. However, there are still uncertainties and limitations in the data and models used in this study. The first limitation is the missing data. In the process of data acquisition, some values are lost because of routine maintenance of the online monitoring network, equipment failure, or manual adjustment of sensor parameters (Chen et al., 2023; Chen and Xue, 2023). Because the model in this study was developed based on monitoring data, missing information may have affected its prediction accuracy. In addition, owing to the limitations of the monitoring equipment and test methods, the collected water quality data contain inherent noise (Kang and Lin, 2007), which may reduce the data quality. The interactions between nitrogen and phosphorus can also lead to errors. These factors may affect the training and prediction processes of the models (Zhu et al., 2023).

Another limitation is that the present study adopts a posterior assimilation strategy, in which the Kalman filter is applied after each model has generated its raw prediction. While this approach is computationally lightweight and model-agnostic, it cannot influence the internal representation learned during training. Exploring whether the Kalman gain can be injected directly into LSTM gating mechanisms, or whether hybrid training schemes (e.g., jointly optimizing network parameters and filter covariances) can be devised, therefore represents a promising direction for future research. Such an end-to-end assimilation framework would shift the filter from a mere post-processing tool to an integral component of the learning algorithm, potentially improving both robustness and interpretability.

6. Conclusions

This study undertook water quality evaluation and identification of key pollutants using daily water quality monitoring data from September 2018 to September 2023 in the Dahei River basin. The objective was to establish characteristics for water quality prediction targets and develop a water quality parameter prediction model utilising LSTM, RF, XGBoost, and SVR based on historical data. Subsequently, a data assimilation method employing KF and the four prediction models were formulated. This method was applied to datasets from different locations and with different temporal resolutions. The main conclusions are as follows:

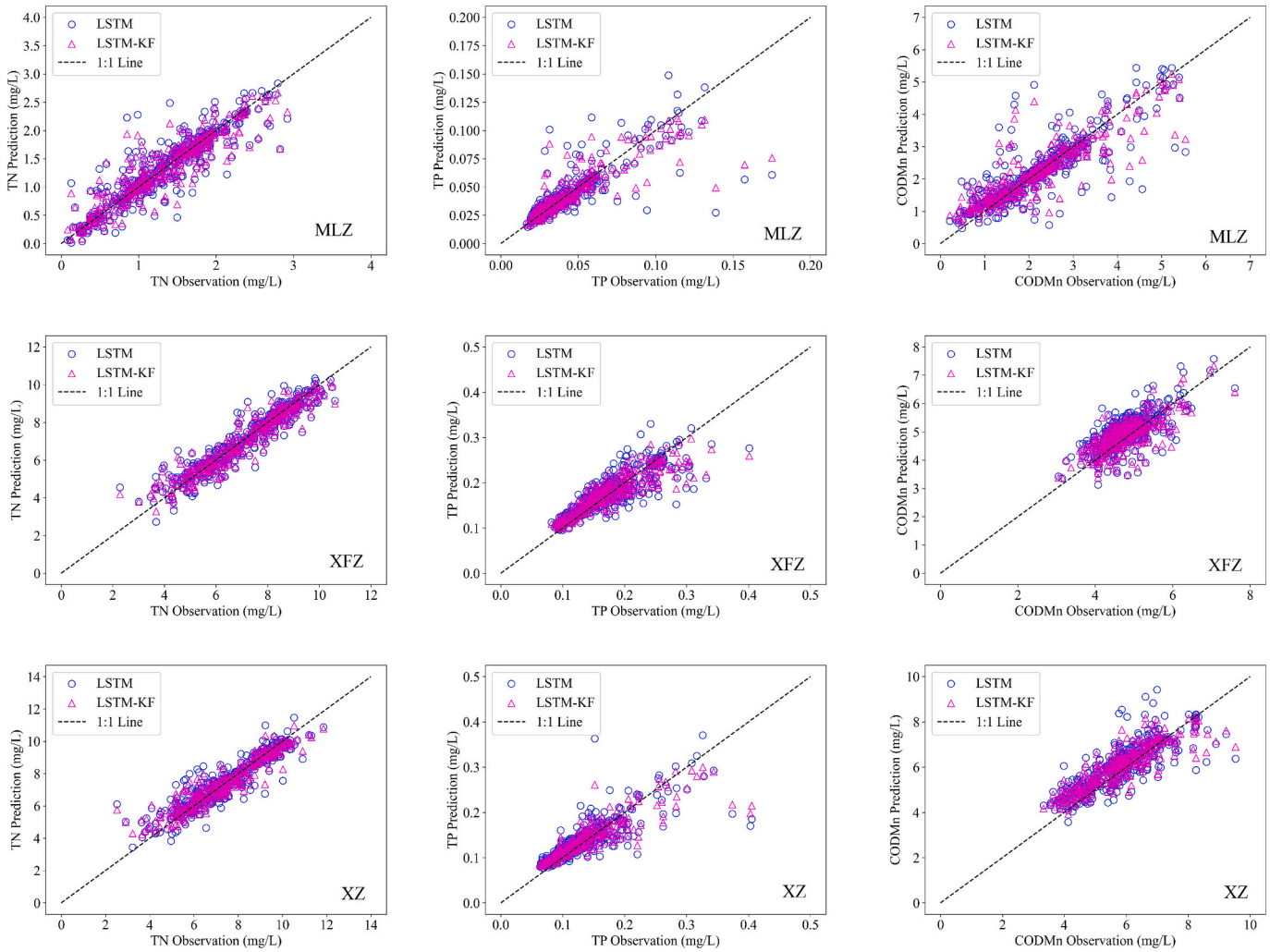


Fig. 9. Scatter plots of predicted and observed values, with Kalman-filtered assimilated data in Xiaoqing River basin.

(1) In this study, three main water pollution indices, TN, TP, and COD_{Mn}, were extracted by evaluating the water quality monitoring data of the Dahei River Basin. The LSTM model exhibited the highest prediction performance for these three indices, surpassing that of the random forest model. The prediction accuracies of XGBoost and SVR were relatively low, and the errors were large. Therefore, when predicting water quality monitoring data, it is recommended to prioritise models such as LSTM, which are more suitable for time-series data.

(2) The Kalman filter data assimilation significantly improved the prediction accuracy of the four models, with the LSTM model exhibiting the highest post-improvement accuracy. The ranking of prediction accuracy for TN was LSTM–KF > RF–KF > SVR–KF > XGBoost–KF. For TP and COD_{Mn}, the ranking was LSTM–KF > RF–KF > XGBoost–KF > SVR–KF. The R^2 values for the LSTM–KF predictions of TN, TP, and COD_{Mn} were 0.909, 0.910, and 0.876, respectively, accompanied by low RMSE values. Relative to other models, this approach offers effective assistance in achieving high-precision predictions of changes in water quality.

(3) The proposed method demonstrates strong applicability across different monitoring sections and datasets with varying resolutions, highlighting its robust portability. This approach offers valuable technical and data support to local ecological and environmental management departments, particularly in semi-arid regions with challenging climatic conditions such as the study area. Moreover, it has the potential to reduce monitoring costs, thereby contributing to sustainable local development.

This lightweight post-prediction assimilation approach can effectively improve prediction accuracy; however, it still presents certain limitations. Future research should explore integrating the Kalman filter with model structures and the training process, aiming to enhance the interaction between data assimilation and model learning. Additionally, the incorporation of spatial data should be considered to improve the model's robustness and its ability to capture both spatial and temporal dynamics.

CRediT authorship contribution statement

Zhenyu Gao: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Guoqiang Wang:** Writing – review & editing, Funding acquisition. **Jinyue Chen:** Writing – review & editing, Funding acquisition. **Lei Fang:** Writing – review & editing. **Shilong Ren:** Writing – review & editing. **A. Yinglan:** Writing – review & editing. **Shuping Ji:** Investigation, Data curation. **Ruobing Liu:** Visualization, Data curation. **Qiao Wang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecoinf.2025.103337.

Data availability

The water quality monitoring data and hydrological data used in this study is under proprietary control can be shared on request. The test data along with the codes used in this study is accessible in Zenodo repository (https://doi.org/10.5281/zenodo.15665429).

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